

Game Theory and Social Choice

Lecture 6: Rationalizability

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Rationality and Solution Concepts

Where We Left Off

Nash equilibrium says each player is playing a best response to what the others are actually doing.

There are two claims here about what is going on in equilibrium.

1. Each player is **rational**,
2. Each player's beliefs about the others are **correct**.

Rationality is a natural assumption, but correctness is weird.

Two Assumptions

If there is only one equilibrium, it's easy to see why we get correctness: they figure it out.

But in games where there are many, how do they know which equilibrium to choose?

Let's see what happens if we drop the correctness assumption and ask what we can conclude just from common knowledge of rationality.

Spoiler

The answer is much less than Nash equilibrium.

What common knowledge of rationality delivers is a weaker notion, **rationalizability**.

Every Nash equilibrium strategy is rationalizable. The converse is not true; many rationalizable outcomes are not equilibria.

Rationalizability

Best Responses

Recall a strategy is a **best response** to a belief about what the other player will do if nothing else does better against that belief.

In the notes we started to iterate this, talking about strategies which are not just best responses, but best responses to best responses, and best responses to best responses to best responses etc.

Let's state that more carefully.

The Definition

Here is the definition, in two parts.

Rationalizability

A **class** of strategies is rationalizable if every member of the class is a best response, assuming the probability that every other player plays some strategy from the class is 1.

A **strategy** is rationalizable if it belongs to some rationalizable class.

The class has to support itself.

Circularity

It looks circular, and in a sense it is, but this is a feature not a bug.

The idea is that you can defend playing s by saying “I played s because I expected them to play t ”.

And then you can be asked to defend expecting t . And you can answer: because they would have expected me to play s' . And so on.

A class is rationalizable when the chain of defences never has to leave the class. Every belief at every step of the chain is justified.

What This Rules Out

Some things, but less than Nash.

A strictly dominated strategy is never a best response to anything, so it is never rationalizable.

In fact, in a two-player game, the rationalizable strategies are exactly the ones that survive iterated elimination of strictly dominated strategies.

Actually, this is literally true but a bit more complicated than it might sound. A strategy counts as dominated if some **mixture** beats it, not just if another pure strategy does.

Every Nash Strategy is Rationalizable

Take any Nash equilibrium. Consider the class consisting of just the strategies in it.

Each is a best response to the others, by the definition of equilibrium. So the class is rationalizable, so each member is rationalizable.

That direction is easy.

A Game With No Pure Equilibrium

Table 1: A game with no pure strategy equilibria

	a	b
A	5, 0	0, 1
B	0, 1	1, 0

No cell has both payoffs bolded, so there is no pure strategy equilibrium.

The Equilibrium

There is exactly one Nash equilibrium.

Row plays $\langle A, \frac{1}{2}; B, \frac{1}{2} \rangle$. Column plays $\langle a, \frac{1}{6}; b, \frac{5}{6} \rangle$.

Row's expected payoff is $\frac{5}{6}$; Column's is $\frac{1}{2}$.

So if rational players play their part of a Nash equilibrium, there is exactly one rational thing for each player to do.

But Everything is Rationalizable

Look at the bolding again. A is a best response to a . And a is a best response to B . And B is a best response to b . And b is a best response to A .

That is a loop, a closed cycle, so $\{A, B\}$ and $\{a, b\}$ form a rationalizable class.

Every strategy in this game is rationalizable, including every mixture. For any strategy you like, there is some opposing strategy it is a best response to.

Two Principles

1. Rational players always play some part of a Nash equilibrium strategy.
2. Rational players always play rationalizable strategies.

On Table 1, Principle 1 rules out all but one strategy per player. Principle 2 says anything goes.

Which is right?

Defending Nash

Repeated Game

If the game is played many times, Principle 1 looks right.

Playing A every time will be a disaster for Row. After a while Column picks up what is going on and responds with b .

In a repeated game, randomising is the right thing to do. But that is a fact about repeated games.

The One-Shot Case

Here is an argument that Principle 1 undermines itself.

Assume the only rational thing to do is play the equilibrium strategy, so Row flips a coin.

Column knows what rationality is, so Column knows Row will flip a coin.

But now Column is facing a decision problem against a known chance device.

Solving that Decision Problem

Against Row's coin flip, every one of Column's options has the same expected return.

So every option is equally rational. So Column can rationally do anything.

Which is exactly what Principle 2 says, and what Principle 1 denied.

Stated Generally

In any game whose only equilibrium is a mixed one, these three cannot all be true.

- A.** Only strategies that are part of Nash equilibria are rational, and all players know this.
- B.** All players are rational, and all players know this.
- C.** Any strategy known to have the same expected return as a rational strategy can be rationally played.

The Derivation

By (A) and (B), the other player will play a mixed strategy.

But then every strategy that you give positive probability to playing has equal expected return. (That's how we find mixed strategy equilibria.)

So by (C) the pure strategies are as rational as the mixture. Which contradicts (A).

Which One Goes?

Conventional wisdom in the game theory texts is that **(C)** fails.

Principle 2 says that **(A)** fails.

This is just a hard question, and for my money the answer to it turns on what we want game theory to **do**. In practice, philosophers often go with Principle 2 and economists go with Principle 1.

Three-Way Games

Why Two Options Hide the Difference

With two choices each, the argument between the Principles comes down to a narrow question.

Is randomising **obligatory**, or is playing a pure strategy from the support **permissible**?

With three choices, the two come apart much further.

A Three-Way Game

Table 2: A three-way game with a poor pure strategy equilibrium

	a	b	c
A	3 , 0	0, 3	0, 2
B	0, 3	3 , 0	0, 2
C	2, 0	2, 0	1 , 1

One cell has both payoffs bolded. Cc is the unique Nash equilibrium, and it pays 1 to each.

The Cycle Again

A is a best response to a . a is a best response to B . B is a best response to b . b is a best response to A .

So A , B , a and b are all rationalizable, even though none of them appears in any equilibrium.

And No Mixed Equilibrium Either

You might expect a mixed equilibrium over those four strategies (i.e., A , B , a and b). But there isn't one.

The only candidate is each player splitting 50/50 between their first two options.

But if the other player is mixing, both players prefer the third option, getting 2, rather than mixing and getting 1.5.

There really is only equilibrium, pure or mixed, but everything is rationalizable.

Now Raise the Stakes

Let's change that equilibrium payout to 2.

Table 3: The same game with the equilibrium raised to 2

	a	b	c
A	3, 0	0, 3	0, 2
B	0, 3	3, 0	0, 2
C	2, 0	2, 0	2, 2

Again Cc is the unique equilibrium, but everything is rationalizable.

Does Your Intuition Move?

In both games, playing A or B is rationalizable and is not part of any equilibrium.

Do you think that what you could plausibly play differs between the games?

Where That Leaves Us

Could it ever be rational, under common knowledge of rationality, to play A or B in either of these games?

If yes, Nash equilibrium is not what rationality requires.

If no, we need to say what rules it out, given that there is an internally coherent justification for it.

Next Time

Everything so far has been simultaneous. Both players move at once, or at least before knowing what the others do.

Next week we let players move in sequence, and see what changes when one player can watch the other first.